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On the deformation of rings and algebras IV

By MURRAY GERSTENHABER[1]

"Jump" deformations of algebras—loosely, those in which any two generic elements in the family of deformed algebras are isomorphic—form a special but important class which includes those deformations in which the generic element is rigid as well as those associated with filtered algebras. The main purpose of this paper is to show that an algebra A which undergoes a jump deformation loses some of its approximate automorphisms ("higher derivations" in the terminology of Jacobson [7]). In the case of characteristic zero this is equivalent to losing some of its moduli. The difficult case is that of characteristic p.

The present work stems from a conjecture of Rim to the effect that if A is a (complete) filtered algebra over a field then the space $H^2(\text{gr } A, \text{gr } A)$ of "infinitesimal deformations" of its (complete) associated graded ring gr A must be strictly larger than the space $H^2(A, A)$ of infinitesimal deformations of A itself. This was proved for A with characteristic zero by Coffee [1], who then showed more generally that if A_t is a jump deformation of A, then the infinitesimal of the deformation from A to A_t can be represented by a 2-cocycle which can be lifted to a 2-coboundary of A_t. Using methods analogous to those of Griffiths [6], we find that if the infinitesimal of the deformation is not zero, then for a filtered algebra A of any characteristic

$$\dim H^2(A, A) < \dim H^2(\text{gr } A, \text{gr } A)$$

as long as the quantity on the right is finite. The idea to translate Griffiths' work into a purely algebraic setting was due to Nijenhuis. While Rim's conjecture as originally stated fails in characteristic p, we show that it can be reformulated to hold there, too.

The theorem that in characteristic zero a jump deformation destroys its own infinitesimal must surely also hold for a jump deformation of a complex manifold X. That is, if X_t describes a differentiable family of manifolds with $X_0 = X$ and all X_t isomorphic for $t \neq 0$, then a cocycle representing the infinitesimal of the deformation should be liftable to a coboundary in a generic X_t. In particular, letting Θ_t denote the sheaf of holomorphic

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tangent vectors on X_t, one should find that $\dim H^1(X, \Theta_t) < \dim H^1(X, \Theta)$ as long as the dimension on the right is finite. This is always the case if X is compact.

In this paper by an "algebra" we mean an algebra A over a commutative unital coefficient ring k which is itself either an algebra over $\mathbf{Q}$ or over the field $\mathbf{F}_p$ of p elements. In the latter case we say that A has characteristic p, and in the former that it has characteristic zero, although it would be more precise to say that A is defined over $\mathbf{Q}$. While k need not be a field we show by example that the main theorem fails if it is defined neither over $\mathbf{Q}$ nor $\mathbf{F}_p$. On the other hand, A need not be associative. It is sufficient that we have a category in which the k-split extensions of A by itself considered as an A-bimodule can be made into a group $H^2(A, A)$. This is the case in the "classical" categories—associative, Lie, and commutative associative algebras—as well as in many others, for example the category of nilpotent associative algebras, to all of which the present paper applies.

1. Basic definitions

We recapitulate here with some minor additions and changes in notation, certain basic definitions and results of [3], [4], and [5].

Let A be a k-algebra, $k[[t]]$ be the ring of formal power series in t over k, and $A[[t]]$ be the $k[[t]]$-module of all power series in t with coefficients in A. By a *deformation* of A we mean a multiplication on $A[[t]]$ of the form

$$f_t(a, b) = ab + tF_1(a, b) + t^2F_2(a, b) + \cdots$$

where the F_i are bilinear functions $A \times A \to A$ ("2-cochains of A with coefficients in itself") which are extended to all of $A[[t]]$ to be $k[[t]]$-bilinear. The algebra so obtained will be denoted A_t and we may speak of a deformation A_t of A with multiplication f_t. Denoting the original multiplication on A by π we write briefly

$$f_t = \pi + tF_1 + t^2F_2 + \cdots .$$

It is required, of course, that A_t be of the same category as A, e.g., associative if A is associative, Lie if A is Lie, and so forth.

A deformation

$$g_t = \pi + tG_1 + t^2G_2 + \cdots$$

is *equivalent* to f_t if there is a "formal linear automorphism" i.e., a series

$$\Phi_t = 1 + t\varphi_1 + t^2\varphi_2 + \cdots$$

where each φ_i is a k-linear map $A \to A$ ($1 = \mathrm{id}_A$), such that

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$$g_t(a, b) = \Phi_t^{-1} f_t(\Phi_t a, \Phi_t b) ,$$

for which we write briefly

$$g_t = f_t \circ \Phi_t .$$

A deformation f_t is *trivial* if it is equivalent to the original multiplication π, i.e., if

$$f_t = \pi \circ \Phi_t$$

for some Φ_t. This implies a fortiori that

$$f_t \equiv \pi \circ \Phi_t \bmod t^n$$

for every $n > 0$. We say that f_t is *quasi-trivial* if for every $n > 0$ there is a $\Phi_{t,n}$ such that

$$f_t \equiv \pi \circ \Phi_{t,n} \bmod t^n .$$

In characteristic zero we shall prove the easy result that a quasi-trivial deformation is trivial.

Recall that if we have a deformation

$$f_t = \pi + tF_1 + t^2F_2 + \cdots ,$$

then the first non-zero F_i is a 2-cocycle of A (with coefficients in itself). The deformation is *restricted* if there is an $n > 0$ such that

$$f_{t^n} = \pi + t^nF_1 + t^{2n}F_2 + \cdots$$

is trivial, in which case we call F_1 a *restricted 2-cocycle*. In characteristic zero, restricted deformations are trivial and restricted 2-cocycles are coboundaries. In characteristic $p > 0$, the least n such that f_{t^n} is trivial is a power $q = p^e$ of the characteristic and any n such that f_{t^n} is trivial is a multiple of q. (For this it is not necessary to assume that k actually has characteristic p but only that in k every integer prime to some prime p is invertible.) Those 2-cocycles F_1 which appear as the linear term of a restricted deformation f_t which becomes trivial when t is replaced by t^{p^e} with a fixed e form an additive subgroup R_eZ^2 (denoted $R_{e-1}Z^2$ in [5]) of the k-module $Z^2 = Z^2(A, A)$ of 2-cocycles of A. We have

$$B^2(\text{2-coboundaries}) = R_0Z^2 \subset R_1Z^2 \subset \cdots .$$

In general R_eZ^2 is not a k-module but only a module over k^{p^e}, the subring of k consisting of all a^{p^e} with $a \in k$. In [2] we viewed the second cohomology group $H^2 = H^2(A, A)$ of A as the space of infinitesimal deformations of A, but if the characteristic is p and k is perfect, i.e. if $k^p = k$, then the classes of elements of RZ^2 should be viewed as inessential. Denoting the submodule

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which they span by RH^2 it is more suggestive for some purposes to view H^2/RH^2 as the space of infinitesimal deformations.

A deformation f_t of A is a *jump deformation* if there is a power series in 2 variables

$$\Phi_{u,t} = 1 + u\varphi_1(t) + u^2\varphi_2(t) + \cdots$$

where each φ_i is a series whose coefficients are k-linear maps $A \to A$ such that

$$f_{(1+u)t} = f_t \circ \Phi_{u,t} .$$

It is easy to verify that if f_t is trivial, quasi-trivial, restricted, or a jump deformation, then the same is true of any equivalent deformation.

A *formal automorphism* ("one parameter family of automorphisms" in [3] or "infinite higher derivation" following Jacobson [7]) of A is a power series

$$\Phi_u = 1 + u\varphi_1 + u^2\varphi_2 + \cdots$$

such that

$$\Phi_u(ab) = (\Phi_u a)(\Phi_u b)$$

or equivalently such that

$$\pi \circ \Phi_u = \pi .$$

A fortiori, for all $n > 0$ we have

$$\pi \circ \Phi_u \equiv \pi \bmod u^{n+1} .$$

If the foregoing holds for some Φ_u and some fixed n then we say that Φ_u is an *approximate automorphism* of order n. Set

$$z(\Phi) = z(\Phi_u) = \varphi_1 \smile \varphi_n + \varphi_2 \smile \varphi_{n-1} + \cdots + \varphi_n \smile \varphi_1 + \delta\varphi_{n+1}$$

(where $(\varphi \smile \psi)(a, b) = (\varphi a)(\psi b)$ and $\delta\varphi(a, b) = a(\varphi b) - \varphi(ab) + (\varphi a)b$), an element of Z^2 which we call the *obstruction cocycle* of Φ. It is easy to see that

$$\pi \circ \Phi_u \equiv \pi + u^{n+1}z(\Phi) \bmod u^{n+2} ,$$

whence $z(\Phi) = 0$ if and only if Φ actually is an approximate automorphism of order $n + 1$. If Ψ is a second approximate automorphism of order n then

$$z(\Phi\Psi) = z(\Phi) + z(\Psi) .$$

The class of $z(\Phi)$ in H^2 depends only on $\Phi \bmod u^{n+1}$. If we consider approximate automorphisms of order n as defined only mod u^{n+1} then for a fixed n these form a group, $\mathrm{Aut}_n A$. There is a morphism $\mathrm{Aut}_{n+1} A \to \mathrm{Aut}_n A$ (reduction mod u^{n+1}) and an element $\Phi \in \mathrm{Aut}_n A$ is in the image if and only if the class of $z(\Phi)$ vanishes. For every integer $m > 0$ there is an "inflation" morphism

$$\mathrm{Aut}_n A \longrightarrow \mathrm{Aut}_{mn+m-1} A$$

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which sends Φ_u to Φ_{u^m}, the latter considered mod $u^{m(n+1)}$.

Let a deformation A_t of A with multiplication f_t be given. We say that an approximate automorphism $\Phi_u = 1 + u\varphi_1 + \cdots + u^n\varphi_n$ of order n of A can be *lifted* if there is an approximate automorphism $\Phi_u(t) = 1 + u\varphi_1(t) + \cdots + u^n\varphi_n(t)$ of f_t (i.e. of A_t) such that $\Phi_u(0) = \Phi_u$. A main result of this paper is that *a jump deformation of A such that every approximate automorphism can be lifted is quasi-trivial.*

2. Some sufficient conditions that a quasi-trivial deformation be trivial

If $\Phi_u = 1 + u\varphi_1 + \cdots + u^{n-1}\varphi_{n-1}$ is an approximate automorphism of order $n-1$ then the first non-zero coefficient is a derivation. Letting this be φ_i, if the characteristic is zero then $\exp u^i\varphi_i$ (where $\exp x = e^x$) is a well-defined formal automorphism and $\Phi_u = (\exp u^i\varphi_i)\Psi_u$ for some approximate automorphism Ψ_u of the form $1 + u^{i+1}\psi_{i+1} + \cdots + u^{n-1}\psi_{n-1}$ also of order $n-1$. Continuing, it follows that in characteristic zero every approximate automorphism of order n is congruent mod u^n to a product of formal automorphisms $(\exp u\theta_1)(\exp u^2\theta_2)\cdots(\exp u^{n-1}\theta_{n-1})$ where the θ_i are derivations.

Suppose now, the characteristic still being zero, that a deformation of the form

$$f_t = \pi + t^nF_n + t^{n+1}F_{n+1} + \cdots$$

is equivalent to another, g_t, whose coefficients vanish up to and including that of t^n. If $g_t = f_t \circ \Phi_t$ then since $f_t \equiv \pi \bmod t^n$ we have in particular that $\pi \circ \Phi_t \equiv \pi \bmod t^n$, so Φ_t is an approximate automorphism of order $n-1$ of the original algebra A. (The use of the parameter t instead of u is only a matter of notation.) Since the characteristic is zero, the obstruction cocycle $z(\Phi)$ is a coboundary, for the preceding remarks show that Φ_t is congruent modulo t^n to a full formal automorphism and therefore *a fortiori* to an approximate automorphism of order n. Now mod t^{n+1} we have

$$\pi \equiv g_t \equiv f_t \circ \Phi_t \equiv (\pi + t^nF_n) \circ \Phi_t \equiv \pi + t^n[z(\Phi) + F_n],$$

so $F_n = -z(\Phi)$, a coboundary. Setting $z(\Phi) = \delta\psi$, and $\Psi_t = 1 + t^n\psi$, we have

$$f_t \circ \Psi_t \equiv \pi + t^n(F_n + \delta\psi) \equiv \pi \bmod t^{n+1}.$$

This shows that, in characteristic zero, if $f_t \equiv \pi \bmod t^n$ and if f_t is equivalent to a $g_t \equiv \pi \bmod t^{n+1}$, then there is a ψ such that $f_t \circ (1 + t^n\psi) \equiv \pi \bmod t^{n+1}$. Suppose now that f_t is quasi-trivial. Then by definition for every n, f_t is equivalent to an $f_{t,n} \equiv \pi \bmod t^n$. It follows that there exist $\psi_1, \psi_2, \cdots$ such that

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$$f_t \circ (1 + t\psi_1) \equiv \bmod t^2$$

$$f_t \circ (1 + t\psi_1)(1 + t^2\psi_2) \equiv \bmod t^3 ,$$

and so forth. But the infinite product $(1 + t\psi_1)(1 + t^2\psi_2) \cdots$ converges formally to a Ψ_t for which we have $f_t \circ \Psi_t = \pi$, proving

THEOREM 2.1. *In characteristic zero, a quasi-trivial deformation is trivial.*

Suppose now that there is a prime p such that in the coefficient ring k every integer prime to p is invertible: in particular, A may be of characteristic p. We recall some basic facts of [5]. If for some derivation φ of A there is a largest integer q for which there exists an approximate automorphism

$$\Phi_t = 1 + t\varphi + t^2\varphi_2 + \cdots + t^{q-1}\varphi_{q-1}$$

of order $q - 1$ with linear term $t\varphi$, then $q = p^e$ for some $e > 0$ called the *exponent* of φ. (If φ has no finite exponent then we set $e = \infty$.) Those φ with exponent $\geqq e$ form a Lie subalgebra $\mathrm{Der}_e A$ of the Lie algebra $\mathrm{Der}\, A$ of all derivations of A, and one has

$$\mathrm{Der}\, A = \mathrm{Der}_1 A \supset \mathrm{Der}_2 A \supset \cdots .$$

(In characteristic p, $\mathrm{Der}_e A$ is a restricted Lie subalgebra of $\mathrm{Der}\, A$.)

Every 2-cocycle F which appears as the obstruction to an approximate automorphism of order $n - 1$ already appears as the obstruction to one of order $q - 1$ where q is the largest power of p dividing n. For fixed e, those F that appear as obstructions to approximate automorphisms of order $\leqq p^e - 1$ form an additive subgroup of Z^2 which by [5] is identical to R_eZ^2 as described in the preceding section. It is also shown there that if the sequence of $\mathrm{Der}_i A$ terminates, i.e. if there is an e such that $\mathrm{Der}_e A = D_{e+1}A = \cdots$, then for every $\varphi \in \mathrm{Der}_e A$ there exists a formal automorphism of A of the form $1 + t\varphi + t^2\varphi_2 + \cdots$.

THEOREM 2.2. *Let the coefficient ring of the k-algebra A have the property that for some prime p every integer prime to p in k is invertible and suppose that the sequence of* $\mathrm{Der}_i A$ *terminates. Then any quasi-trivial deformation of A is trivial.*

Proof. Suppose that we have a deformation $g_t = \pi + t^m G_m + t^{m+1} G_{m+1} + \cdots$ and a series $\Phi_t = 1 + t\varphi_1 + t^2\varphi_2 + \cdots$ such that $g_t \circ \Phi_t \equiv \pi \bmod t^{m+1}$. Then Φ_t is an approximate automorphism of order $m - 1$ and the last congruence shows that $z(\Phi_t) + G_m = 0$. Thus $-G_m$ is the obstruction cocycle

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to some approximate automorphism. If the sequence $\mathrm{Der}_t A$ terminates at $\mathrm{Der}_e A$, then $-G_m = z(\Psi_t)$ for some approximate automorphism Ψ_t of order $q-1$ where q is a divisor of m of the form p^d with $d \leqq e$. Setting $m = nq$ and writing for $\Psi(t)$ for Ψ_t, we then have $g_t \circ \Psi(t^n) \equiv \pi \bmod t^{m+1}$ with $m/n \leqq p^e$. Note that as m goes to infinity so does n, since e is fixed.

Now suppose that f_t is quasi-trivial, so that for every m, f_t is equivalent to some $f_{t,m} \equiv \pi \bmod t^m$. By the foregoing if $f_{t,m}$ is given then $f_{t,m+1}$ may be taken to be of the form $f_{t,m} \circ \Psi(t^n)$ for some Ψ which we denote by Ψ_m and some n with $m/n \leqq p^e$. Writing $f_t = f_{t,1}$ it follows that there is a sequence of series $\Psi_1, \Psi_2, \cdots$ and integers $n_1, n_2, \cdots$ going to infinity such that

$$f_t \circ \Psi_1(t^{n_1})\Psi_2(t^{n_2}) \cdots \Psi_m(t^{n_m}) \equiv \pi \bmod t^{m+1} .$$

The infinite product $\Psi_1(t^{n_1})\Psi_2(t^{n_2}) \cdots$ then converges to a series $\Psi(t)$ such that $f_t \circ \Psi(t) = \pi$, showing that f_t is trivial.

3. The theory of Coffee in characteristic zero

The theory of Coffee [1], based on ideas of Griffiths [5], describes the behavior under deformation of cohomology groups of all dimensions and therefore, for dimension greater than 2, is tacitly limited to the "classical" cases, where those groups are defined. In every case, however, we can define a 1-cocycle of a k-algebra A to be a derivation of A into itself, while a 2-cocycle is essentially an extension $0 \to A \to B \xrightarrow{\mu} A \to 0$ of A by itself considered as an A-bimodule together with a "splitting" $s: A \to B$ such that $s\mu = \mathrm{id}_A$; the associated bilinear map $F: A \times A \to A$ is defined by $F(a, b) = s(ab) - (sa)(sb)$. The coboundary of a 1-cochain $F: A \to A$ is given by $(\delta F)(a, b) = a(Fb) - F(ab) + (Fa)b$ and the coboundary of a zero-cochain F (which is merely an element of A) is given by $(\delta F)(a) = [a, F] = aF - Fa$. Thus $H^1 = H^1(A, A)$ and H^2 are always defined.

Let A_t be a deformation of A with multiplication f_t. We denote by Z_t^n and H_t^n respectively the $k[[t]]$-module of n-cocycles and the nth cohomology group of A_t with coefficients in itself while δ_t denotes the coboundary operator in A_t. Recall that in the classical cases an m-cochain F and n-cochain G of an algebra have a "bracket product" $[F, G]$ with the property that $[G, F] = (-1)^{(m-1)(n-1)}[F, G]$, and that if the multiplication is π then $\delta F = (-1)^{m-1}[\pi, F]$ (cf. [2] for the associative case, [8] for the Lie case). Thus in A_t we have $\delta_t F = (-1)^{m-1}[f_t, F]$.

Now any element $z_t \in Z_t^n$ can be written as a power series $z_t = z_0 + tz_1 + t^2 z_2 + \cdots$ where each z_i is an n-cochain of A which is extended to be multilinear over $k[[t]]$. Reduction mod t gives a morphism $Z_t^n \to Z^n$

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which carries coboundaries to coboundaries and therefore induces a morphism $H_t^n \to H^n$. Denoting the images of Z_t^n and H_t^n by Z_0^n and H_0^n, respectively, the elements of these latter modules will be called "liftable" ("extendable" in [1] and [6]). A liftable $z_0 \in Z^n$ is simply an n-cocycle of A for which there exists a $z_t \in Z_t^n$ of the form $z_t = z_0 + tz_1 + t^2z_2 + \cdots$. Suppose that the epimorphism $Z_t^n \to Z_0^n$ has a section, i.e. that there is a k-submodule Z_1^n of Z_t^n which is carried isomorphically onto Z_0^n. Then it is easy to see that every element of Z_t^n can be written as a power series in t whose coefficients are elements of Z_1^n and therefore that $Z_t^n \cong Z_1^n[[t]]$ as a $k[[t]]$-module. If there is no section, then filtering Z_t^n by the powers of t and denoting the completion of the associated graded module simply by $\operatorname{gr} Z_t^n$ we have $\operatorname{gr} Z_t^n \cong Z_0^n[[t]]$.

For H_t^n the matter is less simple, for if $H_t^n \to H_0^n$ has a section then there is an epimorphism $H_0^n[[t]] \to H_t^n$, but this is an isomorphism if and only if H_t^n has no t-torsion. We can eliminate the t-torsion by adjoining t^{-1} to $k[[t]]$, getting the ring $k((t))$ of formal power series with finitely many negative powers of t, and similarly forming $A((t))$. The multiplication f_t turns this into a $k((t))$-algebra which we denote by $A_{(t)}$. The cohomology $H_{(t)}^n$ of this algebra, which is just $H_t^n \otimes k((t))$, the tensor product taken over $k[[t]]$, has no t-torsion, since if $z_t \in Z_t^n$ has the property that $t^m z_t$ is a coboundary in A_t then z_t is already a coboundary in $A_{(t)}$. Those $z \in Z^n$ which can be lifted to elements of Z_t^n representing t-torsion classes of H_t^n are called "*jump*" cycles and form a k-submodule J^n of Z_0^n, the module of liftable cocycles. One now clearly has $H_{(t)}^n \cong (Z_0^n/J^n)((t))$. In particular, if k is a field, in which case so is $k((t))$, then setting $k((t)) = K$ it follows that $\dim_K H_{(t)}^n = \dim_k (Z_0^n/J^n)$. Note that every coboundary is a jump cocycle since δc can be lifted to $\delta_t c$ and therefore Z_0^n/J is a quotient of H_0^n. If $\dim_k H^n < \infty$ then it follows that $\dim_K H_{(t)}^n$ will be less than $\dim_k H^n$ if either there is a non-liftable n-cocycle or if there is a non-trivial jump n-cocycle, i.e. one which is not a coboundary.

Following Griffiths and Coffee we examine next the obstruction to lifting an arbitrary $z_0 \in Z^n$. Writing $f_t = \pi + tF_1 + t^2F_2 + \cdots$, to lift we must solve

$$[f_t, z_t] = [\pi + tF_1 + t^2F_2 + \cdots, z_0 + tz_1 + t^2z_2 + \cdots] = 0$$

for $z_1, z_2, \cdots$. Collecting powers of t and replacing $[\pi, z_i]$ by $(-1)^{n-1}\delta z_i$, in effect we must solve successively the equations

$$(-1)^n \delta z_1 = [F_1, z_0] ,$$
$$(-1)^n \delta z_2 = [F_2, z_0] + [F_1, z_1], \quad \cdots .$$

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If $z_1, z_2, \cdots, z_m$ have been found satisfying the first m of these, then $[F_{m+1}, z_0] + [F_m, z_1] + \cdots + [F_1, z_m]$ is easily shown to be an $(n+1)$-cocycle whose cohomology class is the obstruction to solving the $m+1^{st}$ equation. Note that $z_0 + tz_1 + \cdots + t^m z_m$ is an n-cocycle of A_t/t^{m+1}. With these notations we have

THEOREM 3.1. *If we have an n-cocycle $z_0 + tz_1 + \cdots + t^m z_m$ of A_t/t^{m+1} then the obstruction to continuing it to one of A_t/t^{m+2} is a jump class in H^{n+1}, and conversely every jump class is such an obstruction.*

Proof. For the direct part, observe that

$$\begin{aligned}&(-1)^{n-1}\delta_t(z_0 + tz_1 + \cdots + t^m z_m)\\ &= [\pi + tF_1 + t^2F_2 + \cdots, z_0 + tz_1 + \cdots + t^m z_m]\\ &= t^{m+1}([F_{m+1}, z_0] + [F_m, z_1] + \cdots + [F_1, z_m]) + \text{higher terms},\end{aligned}$$

showing that the obstruction, which is the class of the term in parentheses, can be lifted to a cocycle which, when multiplied by t^{m+1}, becomes a coboundary. It is therefore a jump class.

Conversely, suppose that $y_0 \in Z^{n+1}$ defines a jump class, i.e. that it has a lifting $y_0 + ty_1 + t^2y_2 + \cdots$ such that

$$\begin{aligned}&t^{m+1}(y_0 + ty_1 + t^2y_2 + \cdots) = \delta_t(b_0 + tc_1 + t^2c_2 + \cdots)\\ &= (-1)^{n-1}[\pi + tF_1 + t^2F_2 + \cdots, c_0 + tc_1 + t^2c_2 + \cdots]\end{aligned}$$

for some m and 1-cochain $c_0 + tc_1 + t^2c_2 + \cdots$. Comparing powers of t shows that $c_0 + tc_1 + \cdots + t^m b_m$ is a cocycle of A_t/t^{m+1}, and the class of $y_0 + ty_1 + \cdots$ is the obstruction to extending this cocycle to one of A_t/t^{m+2}.

From all the foregoing it follows, in particular, that if k is a field, if either Z^n or Z^{n-1} contains a non-liftable element, and if $\dim_k H^n$ is finite, then

$$\dim_{k((t))} H_t^n < \dim_k H^n .$$

Recall that if $f_t = \pi + tF_1 + t^2F_2 + \cdots$ defines a deformation of A then the first non-zero F_i is a 2-cocycle of A [3]. Moreover, if f_t is not trivial, then replacing f_t if necessary by an equivalent deformation for which the first non-vanishing F_i has subscript as large as possible, say m, then F_m is not cohomologous to zero. In the case of characteristic zero we call the class of F_m the "infinitesimal" of the deformation f_t, since we then have the following: If $g_t = \pi + t^n G_n + t^{n+1}G_{n+1} + \cdots$ with $G_n \neq 0$ is equivalent to $f_t = \pi + t^m F_m + t^{m+1}F_{m+1} + \cdots$ then $n \leqq m$; furthermore, if $n < m$ then G_n is cohomologous to zero while if $n = m$ then G_n is cohomologous to F_m. The case of characteristic p is discussed later. But observe that whatever the

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coefficient ring, if we set $f_t^{(n)} = (d/dt)^n f_t$, then $f_t^{(n)}/n!$ is well-defined, since $n!$ formally divides $f_t^{(n)}$. We can therefore define a deformation of the $k[[t]]$-algebra A_t to an algebra over $(k[[t]])[[u]] = k[[t, u]]$ by taking as new multiplication

$$f_{t+u} = f_t + uf_t' + u^2(f_t''/2!) + \cdots . \tag{1}$$

Since the first non-vanishing coefficient of u^i on the right is a 2-cocycle of A_t, we have, regardless of characteristic,

LEMMA 3.1. *If A_t is a deformation of A with multiplication f_t then the first cochain $f_t^{(m)}/m!$ which is not zero is a 2-cocycle of A_t.*

An easy computation shows that if we have an equivalent deformation $g_t = f_t \circ \Phi_t$, then

$$g_t' = f_t' \circ \Phi_t - [g_t, \Phi_t^{-1}\Phi_t'] .$$

The bracket term is the coboundary relative to g_t of $\Phi_t^{-1}\Phi_t'$, so g_t' is cohomologous to $f_t' \circ \Phi_t$. The class of f_t' depends therefore only on the algebra A_t and not on its representation.

Suppose that the characteristic is zero and that f_t is not trivial so that replacing it if necessary by an equivalent deformation we may assume it to be in "reduced" form, i.e. of the form

$$f_t = \pi + t^m F_m + t^{m+1}F_{m+1} + \cdots$$

with F_m not cohomologous to zero. Writing

$$f_t' = mt^{m-1}[F_m + ((m + 1)/m))tF_{m+1} + \cdots] ,$$

one sees that the expression in brackets is a 2-cocycle of A_t congruent to F_m modulo t. Thus F_m can be lifted. Now a cocycle of A which is liftable in one deformation is liftable in any equivalent one, so we have

THEOREM 3.2. *If A_t is a deformation of an algebra A of characteristic zero, then the infinitesimal of the deformation can be lifted.*

Suppose now that A is of characteristic zero and that A_t is a jump deformation of A with multiplication f_t. By definition, $f_{t+tu} = f_t \circ \Phi(t, u)$ for some

$$\Phi(t, u) = 1 + u\mathcal{P}_1(t) + u^2\mathcal{P}_2(t) + \cdots .$$

Replacing u by tu in (1) we see that $tf_t' = \delta_t\mathcal{P}_1(t)$. Thus the class of f_t' is a torsion element of H_t^2 and the class of the infinitesimal of the deformation is therefore a jump class in H^2. We have, therefore, the following extension of Coffee's theorem ([1]):

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THEOREM 3.3. *Let A_t be a non-trivial jump deformation of an algebra A of characteristic zero. Then the infinitesimal of the deformation is a jump class. In particular, if A is defined over a field k of characteristic zero and if* $\dim_k H^2 < \infty$, *then setting* $k((t)) = K$ *we have*

$$\dim_K H_t^2 < \dim_k H^2 .$$

One of the moduli of A has therefore indeed been destroyed by the jump deformation, namely the infinitesimal of the deformation itself! The contrapositive is perhaps more striking:

COROLLARY. *If A is defined over a field k of characteristic zero and if* $\dim_k H^2$ *is finite, then a jump deformation which leaves* $\dim_k H^2$ *unchanged must be trivial.*

Suppose now that $\mathfrak{A}$ is an algebra with a non-negative filtration, $\mathfrak{A} \supset F^0\mathfrak{A} \supset F^1\mathfrak{A} \supset \cdots$, that the filtration is separated, i.e., that $\bigcap F^i\mathfrak{A} = 0$, and that $\mathfrak{A}$ is complete in the filtration topology. Denoting the complete associated graded ring of $\mathfrak{A}$ by A we can by [4] define a deformation $\mathrm{Def}_t A = A_t$ of A with a multiplication $f_t = \pi + tF_1 + t^2F_2 + \cdots$ having the following property: For every $a, b \in A$ the series

$$f_1(a, b) = ab + F_1(a, b) + F_2(a, b) + \cdots$$

converges, so A_1 is well defined, and A_1 is isomorphic as a filtered algebra to $\mathfrak{A}$. Since $1 + u$ is a unit of $k[[t, u]]$, it follows from [4, p. 17], that A_t is a jump deformation of $A_1 = \mathfrak{A}$. Applying the Corollary we obtain the original version of Coffee's theorem:

THEOREM 3.4. *Let $\mathfrak{A}$ be an algebra over a field of characteristic zero which is complete in a separated non-negative filtration, and let A be its complete associated graded algebra. If* $\dim H^2(A, A) = \dim H^2(\mathfrak{A}, \mathfrak{A}) < \infty$ *then $\mathfrak{A}$ is isomorphic to A as a filtered algebra,* i.e., *$\mathfrak{A}$ is graded.*

This proves Rim's conjecture in characteristic zero. The main purpose of what follows is to extend the foregoing results to the case of characteristic $p > 0$. There, a jump deformation may not be quasi-trivial and yet may leave $\dim H^2$ unchanged. This implies that every derivation can be lifted, but, as we shall see, there must exist some higher approximate automorphism which is not liftable.

The fruitful idea of adapting Griffith's work to a purely algebraic setting appears in a manuscript of Nijenhuis and Richardson which unfortunately remains unpublished.

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4. Lifting approximate automorphisms in characteristic p

In the case of characteristic zero it is a particular consequence of the theorems of the preceding section that if A_t is a deformation of A then every derivation of A can be lifted to one of A_t if and only if H_t^2 has no t-torsion. An imprecise analog in characteristic p would be the assertion that "if every approximate automorphism of A of order $\leqq p^e$ can be lifted then $H_t^2/R_eH_t^2$ has no t-torsion and conversely". The direct assertion fails for the technical reason that $R_eH_t^2$ is not a $k[[t]]$-module, while there is only a weak converse asserting the liftability of approximate automorphisms not to A_t but to $A_{t^{1/n}}$ where $n = p^{e(e+1)/2}$ (a result which may not be the best possible). Nevertheless the basic idea in characteristic zero does carry over to characteristic p. Henceforth unless otherwise specified, A_t will be a deformation of a k-algebra A of characteristic p, the multiplication in A_t being of the form $f_t = \pi + tF_1 + t^2F_2 + \cdots$.

LEMMA 4.1. *Let $\Phi_u(t) = 1 + u\varphi_1(t) + \cdots + u^n\varphi_n(t)$ be an approximate automorphism of A_t of order n and suppose that its obstruction cocycle*

$$z(\Phi(t)) = \varphi_1(t) \smile \varphi_n(t) + \cdots + \varphi_n(t) \smile \varphi_1(t)$$

(the cup product being taken relative to the multiplication f_t) is of the form $t^m(G_0 + tG_1 + t^2G_2 + \cdots)$. If every approximate automorphism of A of order $\leqq n + 1$ can be lifted, then there is an approximate automorphism of order n of A whose obstruction is the class of G_0.

Proof. Let the order of $\varphi_i(t)$, ord $\varphi_i(t)$, be the largest integer ν such that t^ν divides $\varphi_i(t)$ and define the "grade" of $\Phi_u(t)$ to be min $[(1/i)$ ord $\varphi_i(t)]$ for $i = 1, \cdots, n$. If this grade is r, define the "index" of $\Phi_u(t)$ to be the least j such that $(1/j)$ ord $\varphi_j(t) = r$. Note that if $\Phi_u(t)$ varies but

$$z(\Phi(t)) = t^m(G_0 + tG_1 + \cdots)$$

where m and G_0 remain fixed, then the grade of Φ_u remains bounded. Now from amongst all approximate automorphisms $\Phi_u(t)$ of A_t whose obstruction cocycle has the same lowest order term t^mG_0, choose those of highest grade. From amongst those of highest grade, choose one of highest index, which we may suppose that $\Phi_u(t) = 1 + u\varphi_1(t) + \cdots + u^n\varphi_n(t)$ denotes. If the grade of $\Phi_u(t)$ is r then we can replace u by $t^{-r}u$ to get $\Psi_u(t) = 1 + u\psi_1(t) + \cdots + u^n\psi_n(t)$ where $\psi_i(t) = t^{-ri}\varphi_i(t)$, which contains no negative powers of t although it may contain fractional ones. Moreover, one has

$$z(\Psi(t)) = t^{m-(n+1)r}(G_0 + tG_1 + \cdots)$$

for some $G_1, G_2, \cdots$, where surely $m - (n + 1)r \geqq 0$. We shall show that

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$m-(n+1)r=0$. For by the definition of the grade, $\Psi_u(0)\neq 1$ since if the index of $\Phi_u(t)$ is j then $\psi_j(0)\neq 0$ and $\Psi_u(0)=1+u^j\psi_j(0)+\cdots+u^n\psi_n(0)$. This is an approximate automorphism of A whose obstruction would vanish were $m-(n+1)r>0$. It could then be extended to an approximate automorphism $1+u^j\psi_j(0)+\cdots+u^n\psi_n(0)+\psi_{n+1}$ of order $n+1$. By [5, Theorem 3], this approximate automorphism can be factored into a product $\Psi'_{u^j}\Psi''_u$ of approximate automorphisms, each of order $n+1$ where

$$\Psi'_{u^j}=1+u^j\psi'_j+u^{2j}\psi'_{2j}+\cdots+u^{lj}\psi'_{lj}$$

is a polynomial in u^j, lj is the largest multiple of j not exceeding $n+1$, and $\psi'_j=\psi_j(0)$. Now $\Psi'_u=1+u\psi'_j+\cdots+u^l\psi'_{lj}$ is an approximate automorphism of A of order $l\leqq n+1$ and by hypothesis can be lifted to an approximate automorphism

$$\Psi'_u(t)=1+u\psi'_j(t)+\cdots+u^l\psi'_{lj}(t)$$

of A_t of order l.

Consider now

$$\Gamma_u(t)=\Psi'_{t^{jr}u^j}(t)=1+u^jt^{jr}\psi'_j(t)+\cdots+u^{lj}t^{ljr}\psi'_{lj}(t)$$

which contains only positive integral powers of t. Since this is a polynomial in u^j, we may view it as an approximate automorphism of order $(l+1)j-1\geqq n+1$. Therefore $f_t\circ\Gamma_u(t)^{-1}\equiv f_t \bmod u^{n+2}$. Thus

$$z\big(\Gamma_u(t)^{-1}\Phi_u(t)\big)=z\big(\Phi_u(t)\big)=t^m(G_0+tG_1+\cdots).$$

But $\Gamma_u(t)^{-1}\Phi_u(t)$ has grade at least r since the grade of a product, as one may readily verify, is greater than or equal to the minimum of the grades of the factors. If the grade of $\Gamma_u(t)^{-1}\Phi_u(t)$ equals r, then its index is greater than j since the coefficient of u^j is $\varphi_j(t)-t^{jr}\psi'_j(t)$ in which the second term cancels the lowest order term of $\varphi_j(t)$. This contradiction to the choice of $\Phi_u(t)$ shows that, for our $\Phi_u(t)=1+u\varphi_1(t)+\cdots+u^n\varphi_n(t)$ of highest grade and index with

$$z\big(\Phi(t)\big)=t^m(G_0+tG_1+\cdots)$$

and m, G_0 fixed, we do indeed have $m=(n+1)r$ where r is the grade. Setting

$$\Psi_u(t)=\Phi_{t^{-r}u}(t)=1+ut^{-r}\varphi_1(t)+\cdots+u^nt^{-nr}\varphi_n(t)$$

we therefore have $z\big(\Psi_n(t)\big)=G_0+tG_1+\cdots$. Therefore $G_0=z\big(\Psi(0)\big)$, i.e., G_0 is the obstruction to an approximate automorphism of order n of A, as asserted.

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We now have the direct part of the imprecise theorem quoted at the outset, where tacitly $R_sH_t^2$ is extended to a $k[[t]]$-module.

THEOREM 4.1. *Suppose that A_t is a deformation of A such that every approximate automorphism of order $\leqq p^s$ of A is liftable to A_t. Let α be an element of H_t^2 such that $t^m\alpha \in R_sH_t^2$ for some $m > 0$. Then $\alpha \in R_sH_t^2$.*

Proof. It is sufficient to show that if a 2-cocycle of A_t of the form $t^m(G_0 + tG_1 + \cdots)$ is in $R_sZ_t^2$ then $\alpha = G_0 + tG_1 + \cdots$ is in $R_sZ_t^2$. Setting $q = p^s$, the preceding implies that there exists a $\Phi_u = 1 + u\varphi_1 + \cdots + u^{q-1}\varphi_{q-1}$ which is an approximate automorphism of order $q - 1$ with $z(\Phi) = G_0$. By hypothesis we can lift $1 + u\varphi_1 + \cdots + u^{q-1}\varphi_{q-1}$ to some approximate automorphism $1 + u\varphi_1(t) + \cdots + u^{q-1}\varphi_{q-1}(t)$ whence setting

$$\Phi_u(t) = 1 + u\varphi_1(t) + \cdots + u^{q-1}\varphi_{q-1}(t)$$

we have $z(\Phi(t)) = G_0 + tG_1' + t^2G_2' + \cdots$ for some $G_1', G_2', \cdots$. Thus $t^{m+1}[(G_1 - G_1') + t(G_2 - G_2') + \cdots] \in R_sZ_t^2$, so $G_1 - G_1' \in R_sZ^2$. By the same argument there exists, therefore, an element of $R_sZ_t^2$ of the form $(G_1 - G_1') + t^2G_2'' + \cdots$. Adding t times this to $G_0 + tG_1' + \cdots$ shows that there is an element of $R_sZ_t^2$ of the form $G_0 + tG_1 + t^2$ (higher terms). Continuing the process indefinitely shows that the original 2-cocycle $G_0 + tG_1 + t^2G_2 + \cdots$ is in $R_sZ_t^2$.

Note that the liftability of all approximate automorphisms of some high order, say n, does not necessarily imply the liftability of those of order $< n$. To illustrate the difficulty, suppose that approximate automorphisms of order 2 are liftable and that we wish to lift an approximate automorphism $1 + u\varphi$ of order one, i.e., in effect we wish to lift the derivation φ. By hypothesis we can lift $1 + u^2\varphi$ considered as an approximate automorphism of order 2, but the lifting will be of the form $1 + u\varphi_1(t) + u^2\varphi_2(t)$ with $\varphi_1(0) = 0$ and $\varphi_2(0) = \varphi$, and not necessarily of the form $1 + u^2\varphi(t)$ with $\varphi(0) = \varphi$. On the other hand, the liftability of all approximate automorphisms of order $\leqq n$ implies, as we show next, the liftability of those of all orders up to but not including the next higher power of p.

LEMMA 4.2. *Let A_t be a deformation of A, let e be any positive integer, and suppose that every approximate automorphism of A of order $\leqq p^{e-1}$ can be lifted to an approximate automorphism of A_t. Then every approximate automorphism of order $\leqq p^e - 1$ is liftable.*

Proof. Suppose not. Let e be the least positive integer such that every approximate automorphism of order $\leqq p^{e-1} = q$ can be lifted but for which there exists some approximate automorphism of order $n \leqq pq - 1$ which

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cannot be lifted. Among these non-liftable approximate automorphisms pick one, say $\Phi = 1 + u^j\varphi_j + \cdots + u^n\varphi_n$, such that the first non-vanishing φ_i has index as large as possible, which here we assume to be j. We shall have a contradiction as soon as we can find any approximate automorphism Γ_t of A_t of order n of the form $1 + u^j\varphi_j(t)$ + higher terms with $\varphi_j(0) = \varphi_j$, for $\Phi\Gamma_0^{-1}$ will then be of the form $1 + u^{j+1}\psi_{j+1} + \cdots$, hence will be liftable, so Φ will be liftable as well. To construct the desired Γ_t, by [5, Theorem 3], we write $\Phi = \Phi'\Phi''$ where each factor is in $\mathrm{Aut}_n A$ and Φ' is of the form $1 + u^j\varphi_j + u^{2j}\varphi'_{2j} + \cdots + u^{lj}\varphi'_{lj}$ with $l = [n/j]$. If $l \leqq q$ then by hypothesis $1 + u\varphi_j + u^2\varphi'_{2j} + \cdots + u^l\varphi'_{lj}$ is liftable to some $1 + u\varphi_j(t) + u^2\varphi'_{2j}(t) + \cdots + u^l\varphi'_{lj}(t)$ and we can set $\Gamma_t = 1 + u^j\varphi_j(t) + u^{2j}\varphi'_{2j}(t) + \cdots + u^{lj}\varphi'_{lj}(t)$. If $l \geqq q$ then by hypothesis $1 + u\varphi_j + u^2\varphi_{2j} + \cdots + u^q\varphi_{qj}$ is liftable to some $1 + u\varphi_j(t) + \cdots + u^q\varphi_{qj}(t)$. But the existence of such an approximate automorphism of A_t shows that $\varphi_j(t) \in \mathrm{Der}_{s-1} A_t$, whence by [5] there is an approximate automorphism of A_t of order $pq - 1$ of the form $1 + u\varphi_j(t) + u^2\gamma_2(t) + \cdots + u^{pq-1}\gamma_{pq-1}(t)$. The desired Γ_t is obtained from this by replacing u by u^j and reducing mod u^n.

To state the following weak converse of Theorem 4.1, denote by $A_{t^{1/n}}$ the algebra obtained by extending coefficients from $k[[t]]$ to $k[[t^{1/n}]]$. The cohomology modules of the new algebra are similarly obtained by extension of coefficients from those of A_t. It follows, in particular, that $R_sH_t^2$ has the property that $t\alpha \in R_sH_t^2$ implies $\alpha \in R_sH_t^2$ if and only if the same is true of $R_sH_{t^{1/n}}^2$. On the other hand, in general there will exist approximate automorphisms $\Phi = 1 + u\varphi_1 + \cdots + u^m\varphi_m$ which cannot be lifted to A_t but which can be lifted to approximate automorphisms $1 + u\varphi_1(t^{1/n}) + \cdots + u^m\varphi_m(t^{1/n})$ of $A_{t^{1/n}}$.

THEOREM 4.2. *Let A_t be a deformation of an algebra A of characteristic p and suppose for some integer $e \geqq 0$ and every non-negative integer $e' \leqq e$ that $t\alpha \in R_{s^{e'}}H_t^2$ implies $\alpha \in R_{s^{e'}}H_t^2$. Then every approximate automorphism of A of order $\leqq p^e$ is liftable to $A_{t^{1/n}}$ where $n = p^{e(e+1)/2}$.*

Proof. If not, the algebras A and A_t being fixed, let e be the least integer for which the conclusion fails. Every approximate automorphism of order $\leqq p^{e-1}$ is then liftable to $A_{t^{1/m}}$ where $m = p^{(e-1)e/2}$, whence by Lemma 4.2, the same is true, setting $q = p^e$, of every approximate automorphism of A of order $q - 1$. Thus by hypothesis there is a non-liftable approximate automorphism $\Phi_u = 1 + u\varphi_1 + \cdots + u^q\varphi_q$ where, however, we can lift $1 + u\varphi + \cdots + u^{q-1}\varphi_{q-1}$ to some $1 + u\varphi_1(t^{1/m}) + \cdots + u^{q-1}\varphi_{q-1}(t^{1/m})$. Letting

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δ_t denote the coboundary operator in A_t and cup-products being understood relative to A_t, it follows that

$$\mathcal{P}_1(t^{1/m}) \smile \mathcal{P}_{q-1}(t^{1/m}) + \cdots + \mathcal{P}_{q-1}(t^{1/m}) \smile \mathcal{P}_1(t^{1/m}) + \delta_t \mathcal{P}_q$$

vanishes at $t = 0$ and therefore is of the form $t^{1/m} z(t^{1/m})$ for some 2-cocycle $z(t^{1/m})$ of $A_{t^{1/m}}$. By hypothesis, $z(t^{1/m}) \in R_e Z^2_{t^{1/m}}$, so $-z(t^{1/m})$ is also the obstruction to some $1 + u\psi_1(t^{1/m}) + \cdots + u^q \psi_q(t^{1/m})$ which is an approximate automorphism of order $q - 1$ of $A_{t^{1/m}}$. Recall that $n = mq$, set

$$\Psi_u(t^{1/n}) = 1 + ut^{1/n}\psi_1(t^{1/m}) + \cdots + u^q t^{q/n} \psi_q(t^{1/m}) ,$$

and set

$$\Phi_u(t^{1/n}) = 1 + u\mathcal{P}_1(t^{1/m}) + \cdots + u^{q-1}\mathcal{P}_{q-1}(t^{1/m}) + u^q \mathcal{P}_q .$$

Then the obstruction to

$$\Gamma_u(t^{1/n}) = \Phi_u(t^{1/n}) \cdot \Psi_u(t^{1/n})$$

is zero, so $\Gamma_u(t^{1/n})$ is actually an approximate automorphism of $A_{t^{1/m}}$ of order q. But $\Psi_u(0) = 1$, so $\Gamma_u(0) = \Phi_u$, i.e., $\Gamma_u(t^{1/n})$ is a lifting of Φ_u, contrary to the hypothesis that no such lifting is possible.

The theorem may not be the best possible in that one may actually be able to lift Φ_u to an approximate automorphism of $A_{t^{1/n'}}$ for some $n' < n = p^{e(e+1)/2}$. The following example shows, however, that we must surely take $n' \geqq q = p^e$: Let k be a field of characteristic p and $K = k(\xi, \eta)$ be an extension with $\xi^{p^{e+1}} = a$, $\eta^{p^e} = b$ with $a, b \in k$ and $\eta^{p^{e-1}} \notin k(\xi)$. Then $K = k(\xi) \otimes_k k(\eta)$, a modular extension of k. This, by Sweedler's theorem [9], is equivalent to K having "enough" approximate automorphisms over k in the sense that every element α of K which is not in k is moved by one. We can deform K to an algebra K_t over $k[[t]]$ in which $\xi^{p^{e+1}} = a$ and $\eta^{p^e} = b + t\xi^{p^e}$. Setting $p^e = q$, we have $K_{t^{1/q}} \cong K[[t]] \otimes k[t^{1/q}]$, the tensor product being over $k[[t]]$. That is, $K_{t^{1/q}}$ is a trivial deformation of K, so the hypothesis of the preceding theorem holds for $K_{t^{1/q}}$ and hence also for K_t. One can readily verify, however, that the deformation $K_{t^{1/n}}$ is not trivial for any $n < q$ and the approximate automorphism Φ_u of order q which is uniquely defined by setting

$$\Phi_u(\xi) = \xi + u\eta, \quad \Phi_u(\eta) = \eta$$

is not liftable to $K_{t^{1/n}}$ for $n < q$ but for $n = q$ is liftable to that $\Phi_u(t^{1/q})$ sending ξ to $\xi + u(\eta - u^{1/q}\xi)$ and sending η to η.

When $e = 0$, Theorem 4.2 asserts that if H^2_t has no t-torsion then every derivation can be lifted, which is already known.

Note that if an approximate automorphism of A can be lifted to $A_{t^{1/n}}$ for some n then it surely can also be lifted to $A_{t^{1/n'}}$ where n' is any multiple

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of n. The theorem suggests that we need be concerned only with the case where n is a power of p. In fact with the previous notations we have

THEOREM 4.3. *If an approximate automorphism Φ_u of A can be lifted to $A_{t^{1/n}}$ for some n then it can also be lifted to $A_{t^{1/q}}$ where q is the highest power of p dividing n.*

Proof. Let $n = mq$ and let ε run through the mth roots of 1. Suppose that $\Phi_u(t^{1/n})$ is a lifting of Φ_u in which the least power $t^{i/n}$ of $t^{1/n}$ which appears, and in which $m \nmid i$, is $t^{j/n}$. Then $[\prod_\varepsilon \Phi_u(\varepsilon t^{1/n})]^{1/m}$ is a well-defined lifting with a higher j. Repeating the process sufficiently often gives a lifting in which only powers of $t^{1/q}$ appear.

5. The quasi-triviality theorem

Let $f_t = \pi + tF_1 + t^2F_2 + \cdots$ define a deformation A_t of an algebra A of characteristic p. If f_t is actually a power series in t^p then $f_t' = (d/dt)f_t = 0$. More generally, let $q = p^e$ be the highest power of p such that f_t can be written as a power series in t^q; we call e the *exponent* of f_t. Then $f_{t+u} = \sum_{i=0}^{\infty} u^i f_t^{(i)}/i!$ actually takes the form

$$f_{t+u} = f_t + u^q f_t^{(q)}/q! + u^{2q} f_t^{(2q)}/(2q)! + \cdots$$

with $f_t^{(q)}/q! \neq 0$. It follows that $f_t^{(q)}/q!$ is a 2-cocycle of A_t. Let the least i with $F_i \neq 0$ and $p^{e+1} \nmid i$ be mq; we shall call this the *index* of f_t. Clearly $p \nmid m$ and since $(d/dt)^q/q!$ applied to t^q gives 1 we have

$$f_t^{(q)}/q! = mt^{(m-1)q}F_{mq} + (m+1)t^{mq}F_{(m+1)q} + \cdots .$$

Dividing by $mt^{(m-1)q}$ and setting $t = 0$ shows that F_{mq} is a 2-cocycle of A and that it is liftable to Z_t^2; we call it the *primary cocycle* of f_t. With these notations we can state the main result.

THEOREM 5.1. *Let $f_t = \pi + tF_1 + t^2F_2 + \cdots$ define a deformation A_t of a k-algebra A of characteristic p and suppose that f_t has exponent e and index mq, where $q = p^e$ and $p \nmid m$. Then:*

1. *The primary cocycle F_{mq} of f_t can be lifted to a cocycle $z_t \in Z_t^2$.*

2. *If f_t is a jump deformation of A then its primary cocycle F_{mq} can be lifted to a $z_t \in Z_t^2$ such that $t^{mq}z_t \in R_eZ_t^2$.*

3. *If f_t is a jump deformation of A and if every approximate automorphism of A can be lifted to A_t then f_t is quasi-trivial.*

Proof. That 1. holds has been shown.

As for 2., suppose that $f_t = \pi + t^qF_q + t^{2q}F_{2q} + \cdots$ is a jump deformation and that $f_{t+tu} = t_t \circ \Theta(t, u)$ where $\Theta(t, u) = 1 + u\theta_1(t) + u^2\theta_2(t) + \cdots$. Since $f_{t+tu} = f_t + u^q(t^q f_t^{(q)}/q!) + \cdots$ it follows that

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$$t^q f^{(q)}/q! = mt^{mq}F_{mq} + (m+1)t^{(m+1)q}F_{(m+1)q} + \cdots$$

is in $R_e Z_t^2$, so we can take

$$z_t = F_{mq} + m^{-1}(m+1)t^q F_{(m+1)q} + \cdots ,$$

proving 2. Note that division by m is permissible since although $R_e Z_t^2$ is not generally a $k[[t]]$-module it surely is a module over the field $\mathbf{F}_p$ of p elements.

As for 3., it is sufficient to show that f_t is equivalent to a deformation $f_t \circ \Phi_t$ of exponent $\geqq e+1$. The index of f_t being mq, we prove the following more precise assertion by induction on the exponent e: there is a formal linear automorphism Φ_t of A such that $\Phi_t \equiv 1 \bmod t^m$, $f_t \circ \Phi_t \equiv f_t \bmod t^{mq}$, and $f_t \circ \Phi_t$ has exponent $\geqq e+1$. One may start the induction formally with $e = -1$ since taking $\Phi_t = 1$ satisfies the assertions. We may suppose that the assertion is true for all exponents $\leqq e-1$ and prove it for exponent e.

Since by hypothesis every approximate automorphism can be lifted, by Lemma 4.1 there is an approximate automorphism $\Phi \in \mathrm{Aut}_{mq-1} A$ with $z(\Phi) = F_{mq}$. Since $p \nmid m$, it follows that $F_{mq} \in R_e Z^2$, whence also $-F_{mq} \in R_e Z^2$. That is, there is an approximate automorphism $\Psi_u = 1 + u\psi_1 + \cdots + u^q\psi_q$ with $z(\Psi) = F_{mq}$. By hypothesis, $\Psi_u \bmod u^q$ can be lifted to an approximate automorphism $1 + u\psi_1(t) + \cdots + u^{q-1}\psi_{q-1}(t)$. Set

$$\Psi_u(t) = 1 + u\psi_1(t) + \cdots + u^{q-1}\psi_{q-1}(t) + u^q\psi_q .$$

Then $\Psi_{u^m}(t)$ is an approximate automorphism of A_t of order $mq-1$. We have

$$f_t \circ \Psi_{u^m}(t) = f_t + u^{mq}[-F_{mq} + tG_1(t)] + u^{mq+1}G_2(t, u)$$

where G_1 and G_2 are power series in t and t, u, respectively. Now in the foregoing equation set $u = t$ and write $\Psi_{t^m}(t) = \Gamma_t$. Then $f_t \circ \Gamma_t \equiv f_t \bmod t^{mq}$, but the coefficient of t^{mq} in $f_t \circ \Gamma_t$ vanishes. If it should happen that the exponent of $f_t \circ \Gamma_t$ is greater than e then taking $\Phi_t = \Gamma_t$ the induction is concluded. We suppose, therefore that the index of $f_t \circ \Gamma_t$ does not exceed e. It may in fact be less.

Now set $e = e_1$, $q_1 = p^{e_1}$, $m = m_1$, $f_t = f_t^1$, $\Gamma_t = \Gamma_t^1$, and $f_t^2 = f_t \circ \Gamma_t^1$. The index of f_t^2 is strictly greater than that of f_t^1, i.e. than $m_1 q_1$ and can be written in the form $m_2 q_2$ with $q_2 = p^{e_2}$ where e_2 is the exponent of f_t^2 and $p \nmid m_2$. By assumption we have $e_2 \leqq e_1$ whence $m_2 > m_1$. If $e_2 < e_1$ then by the inductive hypothesis there is a $\Phi_t^1 \equiv 1 \bmod t^{m_2}$ such that $f_t \circ \Gamma_t^1 \Phi_t^1$ has exponent e_1 where Φ_t^1 is an approximate automorphism of order $m_2 q_2 - 1 \geqq m_1 q_1 - 1$ relative to the multiplication $f_t \circ \Gamma_t^1 \Phi_t^1$. Replacing Γ_t^1 by $\Gamma_t^1 \Phi_t^1$ we may therefore assume in fact that Γ_t^1 can be so chosen that $e_2 = e_1$. Apply-

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ing the same argument to $f_t^2 = f_t^1 \circ \Gamma_t^1$ and so forth, we see that there is a sequence $\Gamma_t^1, \Gamma_t^2, \cdots$ of formal linear automorphisms such that i) $\Gamma_t^i \equiv 1 \bmod t^{m_i}$ with $m_1 < m_2 < \cdots$ and ii) setting $f_t^{i+1} = f_t \circ \Gamma_t^1 \Gamma_t^2 \cdots \Gamma_t^i$ we have $f_t^i \equiv f_t \bmod t^{m_q}$ for all i, iii) the index of f_t^i is monotone increasing, and iv) the exponent of f_t^i either remains constantly equal to e or for some i exceeds e, at which point we stop, setting $\Phi_t = \Gamma_t^1 \cdots \Gamma_t^i$. If the latter case never occurs then by i) the product $\Gamma_t^1 \Gamma_t^2 \cdots$ converges and we take this infinite product to be Φ_t. By ii) we have $f_t \circ \Phi_t \equiv f_t \bmod t^{m_q}$ while iii) and iv) together imply that the exponent of $f_t \circ \Phi_t$ exceeds e. This completes the induction and the proof.

A simple example shows that the foregoing theorem fails if the coefficient ring k is not of characteristic p but has only the property that every integer prime to p is invertible. Let $p = 2$ and $k = Z/4$ (integers mod 4), and let $A = k[X]/X^2$. Denoting the image of X in A by x we have $x^2 = 0$. If we define $F_2 \colon A \times A \to A$ by $F_2(1, x) = F_2(x, 1) = F_2(1, 1) = 0$, $F_2(x, x) = 1$ then $f_t = \pi + t^2 F_2$ defines a deformation of A which if viewed as a multiplication on the $k[[t]]$-module $A[[t]]$ gives an algebra isomorphic to $k[[t]][X]/(X^2 - t^2)$. This is moreover a jump deformation since the latter algebra is isomorphic to $k[[t]][X]/(X^2 - (1 + u)^2 t^2)$ under an isomorphism making x of the former correspond to $(1 + u)x$ of the latter. Therefore setting $\varphi(x) = x$ and $\Phi_u = 1 + u\varphi$ we have $f_t \circ \Phi_t = f_{t+ut}$. (It is accidental that f_t and Φ_u are both polynomials and that φ does not actually involve t.) Now there is besides the zero derivation of A only one other, namely that θ defined by $\theta x = \alpha x$, and this is itself already a derivation of f_t, hence liftable. Moreover, $1 + u\theta$ is an automorphism of both A and the deformed algebra A_t. It is easy to verify that every approximate automorphism of A is a product of ones of the form $1 + u^n\theta$ for various n, so all are liftable. (In fact all are actually formal automorphisms, their liftings are themselves and remain formal automorphisms.) But the deformation defined by f_t is neither trivial nor quasi-trivial because there is no power series of the form $\xi = x + a_1 t + a_2 t^2 + \cdots$ with coefficients $a_1, a_2, \cdots$ in A such that with the multiplication in A we have $\xi^2 \equiv t^2 \bmod t^3$. Thus the theorem fails for A. The difficulty is that the cohomology of A has torsion, specifically, $2F_2 = \delta\varphi$ while F_2 is not itself a coboundary. We could, of course, have started with $k = Z/p^2$ for any prime p.

We note in closing that the main conclusion of Theorem 5.1, namely, that a jump deformation for which all approximate automorphisms are liftable is quasi-trivial, depends only on the multiplication in the deformed

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algebra A_t and not on its representation by f_t. The statement can, therefore, be carried over to more general algebraic deformation theories. The technique of proof however depends on the representation and uses non-invariant concepts[1] like the primary cocycle, a defect which will probably have to be remedied if the theory is to be extended.

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[1] The correct invariant algebraic definition of a deformation stems from the works of S. Lichtenbaum and M. Schlessinger, The cotangent complex of a morphism, Trans. Amer. Math. Soc. **128** (1967), 41–70 and M. Schlessinger, Functors of Artin rings, ibid, **130** (1968), 208–222. We have not used it here because our computations were easier in non-invariant form. However, the Lichtenbaum-Schlessinger definition is basic for any more general theory.

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